

Ashkenazi mutations, DNA testing came into use. Many geneticists prefer to conduct both types of tests, especially for non-Jews. They find DNA testing useful for its simplicity and exceedingly low error rate, but also recommend enzyme testing to guard against the involvement of a previously undetected mutation that would be missed by the mutation-specific DNA tests.

Future Prospects

In the 1990s laboratories in the United States, Canada, and France developed mouse models of Tay-Sachs disease, Sandhoff disease and G_{M2} activator deficiency. These investigations led to a much better understanding of the brain pathology and progression of the diseases. A significant outcome has been the use of the mouse models to experiment with approaches to therapy. A promising approach is based on partially blocking the synthesis of gangliosides with drugs so that accumulation of G_{M2} ganglioside becomes minimal. In addition to “substrate deprivation,” as this blocking action is called, other laboratories are trying gene therapy and drug-based methods for bypassing the Tay-Sachs defect. The combination of carrier testing and prenatal diagnosis to assure the birth of healthy babies, and the more recent prospects for treating affected patients are major advances since the discovery of a cherry-red spot described in the first infant known to have been born with Tay-Sachs disease. SEE ALSO CELL, EUKARYOTIC; DISEASE, GENETICS OF FOUNDER EFFECT; GENE; GENETICS OF DISEASE; HETEROZY-GOTE ADVANTAGE; METABOLIC DISEASE; MUTATION; POPULATION SCREENING; PRENATAL DIAGNOSIS; PROTEINS; RNA PROCESSING; RODENT MODELS.

Roy A. Gravel

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Technical Writer

A technical writer (sometimes called a technical communicator) designs, writes, edits, and produces documents for scientific, technical, industrial, and government organizations. These documents can include technical reports, specifications, reference manuals, operating instructions, policies and procedures, proposals, presentations, brochures, and Web pages.

Goals and Skills Required

The main aim of a technical writer is to communicate scientific and technical information to other people using easily understandable language. To be a technical writer, a person needs strong language skills, demonstrated by college-level training. A college degree in English, journalism, or communication is preferred, according to the U.S. Department of Labor. The

person also should also have some familiarity with scientific or technical topics. Finally, the person should have experience using word processing and desktop publishing software, graphics programs, and Web publishing tools.

Specially trained people began to be employed as technical writers in the late 1930s. Prior to the 1980s, however, most technical documents still were written by scientists, engineers, and other specialists, many of whom found it difficult to write for nontechnical audiences. With the rapid expansion of science and technology, however, the need increased for people who could both understand complex ideas and convey them effectively to a variety of audiences.

Challenges, Advantages, and Drawbacks

Some challenges need to be considered when thinking about a career as a technical writer. For example, a person may have to invest considerable time and money to acquire the knowledge and skills needed. Also, it can be difficult to gain entry-level experience. Technical writing is typically a sedentary profession that does not involve travel. At the same time, it is a demanding profession that can take time and energy away from other, more creative writing pursuits. Working for a company with an established set of document guidelines can be frustrating, and the profession is sometimes criticized for being dry and unimaginative.

Generally, however, the outlook for technical writers is bright. Technical writing is a job growth area: More jobs are being created than are being filled, particularly in the high technology industry. Once employed, a technical writer works on a wide variety of projects, many of which represent the cutting edge of science and technology. The field is supportive of female professionals; more than half of all technical writers are women. While a majority of technical writers are between the ages of twenty-five and forty-four, about 20 percent are over fifty-five years.

Entering the Profession

In addition, technical writing is a profession that pays well. According to a 2000 salary survey by the Society for Technical Communication, the average salary for a technical writer in the United States is about \$52,000. An entry-level technical writer makes about \$37,000, which compares favorably with entry-level positions in other fields. The average salary for a senior-level technical writer with supervisory responsibilities is about \$65,000. Salary level also depends on geographic location, level of education, and years of experience in the technical writing field. People interested in seeking employment as technical writers should pursue volunteer and internship opportunities, develop a portfolio of their work to show potential employers, check classified advertisements and company Web sites for job openings, write directly to personnel departments, and/or sign up with a job placement agency that specializes in information technology. **SEE ALSO** SCIENCE WRITER.

Cindy T. Christen

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Telomere

eukaryotes organisms with cells possessing a nucleus

nucleases enzymes that cut DNA or RNA

Telomeres are structures found at the ends of chromosomes in the cells of **eukaryotes**. Telomeres function by protecting chromosome ends from recombination, fusion to other chromosomes, or degradation by **nucleases**. They permit cells to distinguish between random DNA breaks and chromosome ends. They also play a significant role in determining the number of times that a normal cell can divide. Unicellular forms whose cells have no true nuclei (prokaryotes) possess circular chromosomes that, therefore, have no ends. Thus, prokaryotes can have no telomeres.

Structure

Telomeres are extensions of the linear, double-stranded DNA molecules of which chromosomes are composed, and are found at each end of both of the chromosomal strands. Thus, one chromosome will have four telomeric tips. In humans, the forty-six chromosomes are tipped with ninety-two telomeric ends.

In most eukaryotic forms, telomeres consist of several thousand repeats of the specific nucleotide sequence TTAGGG and occur in organisms ranging from slime molds to humans. The entire length of repeated telomere sequences is known as the terminal restriction fragment (TRF). Sequences different from TTAGGG are found in more primitive eukaryotic forms, such as the ciliated protozoan *Tetrahymena*, in which Elizabeth Blackburn first characterized the repeated telomere sequence.

The **polymerases** that copy the chromosomes of DNA strands are unable to copy completely to the end. This became known as the "end-replication problem" when it was first recognized in the late 1960s. The TRF acts like a **buffer** that protects the information-containing genes, so that the loss of some telomeric nucleotide sequences at each round of DNA replication does not result in the loss of genetic information. The telomeres themselves end in large duplex loops, called T-loops.

A Simple Counting Mechanism

For the first half of the twentieth century it was believed that cells cultured in laboratory glassware could replicate indefinitely if the correct nutrient **media** and other conditions of growth could be found. Repeated initial failure at culturing indefinitely replicating cells was followed by success in the late 1940s, when the immortal L929 cancer cell population was developed from mouse tissue. Later, other immortal cell populations were found, including the first human cell line, HeLa, derived from a human cervical carcinoma.

polymerases enzyme complexes that synthesize DNA or RNA from individual nucleotides

buffers substances that counteract rapid or wide pH changes in a solution

media (bacteria) nutrient source

somatic nonreproductive; not an egg or sperm

stem cell cell capable of differentiating into multiple other cell types

germ cells cells creating eggs or sperm

fibroblasts undifferentiated cells that normally give rise to connective tissue cells

Using what is called the TRAP assay (telomeric repeat amplification protocol), it has been found that about 90 percent of all human tumors produce telomerase, whereas the only normal adult **somatic** cells that produce telomerase are **stem cell** populations found, for example, in skin, the hematopoietic system, **germ cells**, and gut epithelia. In fact, the presence or absence of telomerase is the most specific property that distinguishes cancer cells from normal cells. This difference is currently under investigation as a diagnostic tool. If a chemical could be found to interfere with telomerase activity in cancer cells, an effective control of this disease might be found. Several candidate substances have been identified and are undergoing extensive studies in animals.

Telomerase is switched on in virtually all human cells at the moment of conception, but as the embryo matures the telomerase becomes repressed in all but the germ cells and stem cell populations. Further, the level of telomerase expressed in stem cells is much less than that expressed in cancer cells. Interestingly, telomerase expression has been found to occur in all the cells of animals that age slowly or not at all. These are animals, such as the American lobster and the rainbow trout, that do not stabilize at a fixed size in adulthood.

On the human genome, an enzyme known as human telomerase reverse transcriptase (hTERT) is found on the most distal gene on chromosome 5p. The transfection (introduction) of hTERT into cultured normal human **fibroblasts** has resulted in telomere elongation, telomerase expression, and the immortalization of these otherwise mortal cells. After several hundred population doublings, the transfected cells exhibit some drift from the diploid number of chromosomes but cancer cell properties do not occur. This experiment proves that telomerase is not a cancer enzyme but an immortalization enzyme. The ability to immortalize normal human cells via hTERT has important potential applications. Some immortalized cells could be cultured in the lab to produce therapeutically useful molecules. Others might be used directly within the body to repair tissue or replace lost or damaged cells. **SEE ALSO** CHROMOSOME, EUKARYOTIC; DNA POLYMERASES; REPLICATION; REVERSE TRANSCRIPTASE.

Leonard Hayflick

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Transcription

Transcription is the process in which genetic information stored in a strand of DNA is copied into a strand of RNA. The sequence of the four bases in DNA, which are adenine (A), cytosine (C), guanine (G), and thymine (T), is preserved in the sequence of the four bases in RNA, which are A, C, G, and uracil (U).

Functions of RNA Transcripts

RNA molecules have various functions in the cell. Many of the functions are associated with translation, in which the genetic code of messenger RNA molecules is used to help the **ribosomes** synthesize a specific protein. In addition, ribosomal RNA is the main component of the ribosome, and transfer RNA does the actual translating from nucleotide sequence into amino acid sequence.

RNA molecules may also function as enzymes. They do so either alone or in association with proteins. RNA molecules associate with proteins, for example, when they serve as components of machinery that helps make other, newly formed RNA molecules functional.

RNA is chemically better suited to carry out certain tasks than is DNA. There are also other reasons RNA, not DNA, is used for these tasks. First, it is desirable to keep DNA available for replication and not tied up with other functions. Second, the small number of DNA molecules in the cell is often insufficient. Creating many identical RNA molecules that are copies of a single segment of DNA provides the necessary numbers. Third, RNA can be differentially degraded when it is no longer needed, providing an important regulatory mechanism that would be unavailable if there were only one type of nucleic acid.

Promoters

Transcription is initiated at regions of DNA called promoters, which are typically 20 to 150 **base pairs** long, depending on the organism. The sequence of bases at a promoter is recognized by RNA polymerase, the enzyme that synthesizes RNA.

The RNA polymerases in bacteria, as well as in viruses in bacteria, are able to recognize particular promoter sequences without the help of any other cellular proteins. However, in eukaryotes and Archaea, other proteins, called initiation factors, recognize the promoter sequence, “recruit” RNA polymerase and other proteins, help the RNA polymerase bind to the DNA, and regulate the enzyme’s activity.

RNA polymerase is assembled on promoters in a particular orientation (Figure 1A). This allows RNA synthesis to start at a precise location and proceed in only one direction, “downstream” toward the gene (Figure 1B).

RNA synthesis

RNA, like DNA, is a **polymer** of nucleotides. Each nucleotide consists of a sugar that is attached to a phosphate group and any one of four bases. The RNA polymerase, as it builds the chain of nucleotides, processes only one of the two **complementary** strands of DNA. This DNA strand is referred to as the template strand. The least confusing name for the other DNA strand is “the nontemplate strand.”

The bases in the newly synthesized RNA are complementary to the bases in the template DNA strand and, therefore, identical in sequence to the bases in the nontemplate strand, except that the RNA contains U where the nontemplate strand of DNA contains T.

ribosomes protein-RNA complexes at which protein synthesis occurs

base pairs two nucleotides (either DNA or RNA) linked by weak bonds

polymer molecule composed of many similar parts

complementary matching opposite, like hand and glove



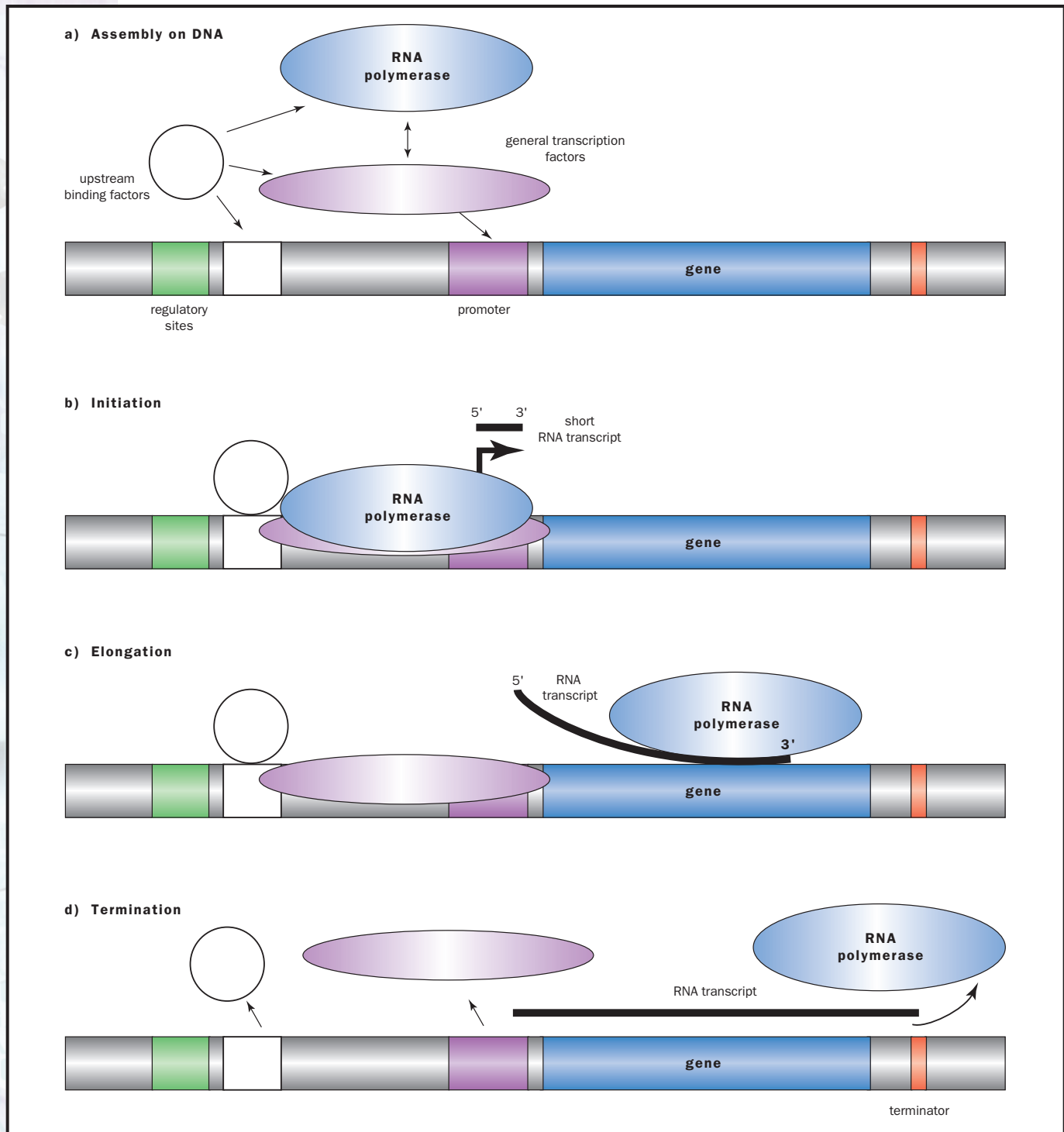
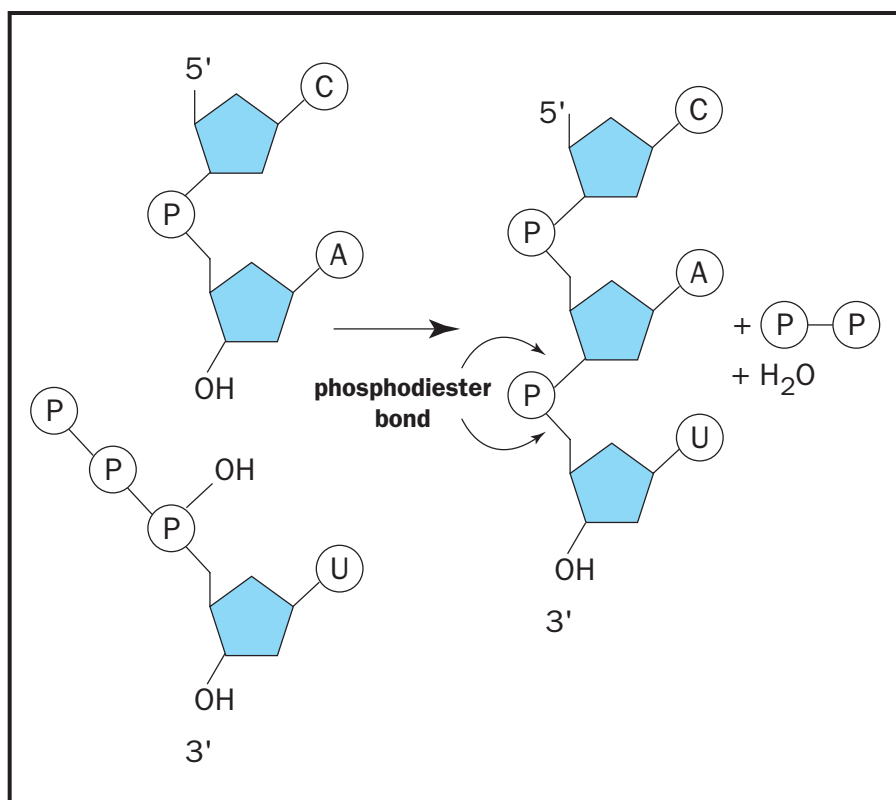


Figure 1. Various steps in the process of transcription. (A) Assembly of the initiation complex. Depending on the conditions RNA polymerase can either bind free general transcription factors and then the promoter, or the general transcription factors that are already promoter-bound themselves. In either case, the upstream binding factors facilitate the interaction of RNA polymerase with the promoter. Note that only one of the upstream regulatory sites is depicted as having a regulatory factor bound. (B) Initiation of RNA synthesis. At precise locations as determined by the promoter DNA sequence, the first and second RNA bases bind to the complex and RNA polymerase catalyzes formation of a covalent bond between them (thick line represents a short, released, abortive transcript). The vertical part of the arrow indicates the position on DNA where RNA synthesis commences, the horizontal part the direction of movement of RNA polymerase on the DNA. (C) Elongation. After promoter clearance, the RNA polymerase extends the growing RNA chain across the gene (the thick line is the growing RNA transcript). (D) Termination. When the elongating complex reaches the terminator, RNA synthesis stops and the complex is released from the DNA.



RNA nucleotides link together by dehydration, the removal of a water molecule and the bonding of the remaining atoms. The pair of P-O-C linkages connecting the two sugars is called a phosphodiester bond. The triphosphate of the incoming nucleotide provides the energy needed to carry out this reaction.

Before the nucleotides are linked together, they exist separately as ribonucleoside triphosphates (NTPs). As shown below, the NTPs contain one of the four common RNA bases, A, C, G, and U, linked to a five-carbon ribose sugar, linked, in turn, to a chain of three phosphate groups. During RNA synthesis, a covalent, “phosphodiester” bond is formed between one of the three phosphate groups on one NTP and a hydroxyl group on another. The two other phosphate groups that were part of the original NTP are released.

RNA synthesis is said to proceed in the 5' to 3' direction, reflecting the fact that the attachment of new nucleotides always occurs at the 3' hydroxyl group of the growing RNA chain. RNA synthesis goes through phases that are typical of **polymerization** processes: initiation, elongation, and termination, yielding an RNA product of defined size and sequence.

Initiation. The first phase of RNA synthesis is initiation (Figure 1B). Initiation starts when the first phosphodiester bond is formed. At precise locations, determined by the promotor DNA sequence, the first and second RNA bases bind to the complex, and RNA polymerase catalyzes the formation of a covalent bond between them.

When the growing RNA chain reaches a length of about ten nucleotides, the complex loses contact with the promoter and starts moving along the DNA. This is referred to as promoter “clearance” or “escape.”

Only a fraction of initiation events lead to promoter clearance. In many instances, an “abortive” RNA molecule, shorter than ten nucleotides, is released from the RNA polymerase, and RNA synthesis begins all over again. Such an abortive molecule is shown in the figure as a thick line.

polymerization linking together of similar parts to form a polymer



Once the growing RNA chain has reached the critical length of about ten nucleotides, the initiation stage is considered to have ended, and elongation begins. In eukaryotes, the transition from initiation to elongation can be triggered by enzymes called kinases, which attach phosphate groups to RNA polymerase, facilitating promoter clearance.

Elongation. Genes range in length from about 80 base pairs of DNA, as is the case for those transcribed into transfer RNA, to more than 1 million base pairs, as is the case for those encoding very long proteins. An RNA polymerase molecule that has disengaged from DNA during elongation would be unable to finish synthesizing the RNA molecule. Thus the enzyme has to traverse even the longest genes (Figure 1C), without falling off.

Along the way, there are DNA sequences that the RNA polymerase traverses considerably more slowly than at its usual rate of about 50 nucleotides per second. At regions called pause sites, it may take longer than 1 second for a single nucleotide to be added to the growing polymer.

In eukaryotes, many genes contain blocks of DNA called **introns**, which disrupt the coding information of the gene. Introns are removed from the newly made RNA by a process called splicing. It is thought that the proteins which carry out the splicing are carried by the RNA polymerase as it is transcribing the gene, allowing the processing of the RNA to occur at the same time as the RNA molecule is synthesized.

Termination. When the RNA polymerase reaches a specific DNA sequence known as a terminator, it slows down and the transcription complex dissociates from the DNA, as shown in Figure 1D. The released RNA polymerase is then free to participate in a new initiation event.

At some terminators, primarily in bacteria, the RNA polymerase is able to respond to the release signal without being helped by any other proteins. Such sites are called intrinsic terminators. At other sites, termination is accomplished only with the aid of additional proteins. These proteins, called termination factors, are also instrumental in causing RNA to be released from the transcribing complex.

“Factor-dependent” terminators have been found in organisms from each of the three domains of life, the eukaryotes, bacteria, and **Archaea**. In eukaryotes, but usually not in bacteria, transcription of most genes proceeds past the end of the gene, as shown in Figure 1D.

The initial RNA molecules are often referred to as “primary” transcripts. In many instances, the primary transcripts must be processed to yield functional, or “mature,” RNA. The processing can involve shortening them by removing their terminal or internal regions, or modifying specific nucleotides in other ways.

Regulation of Transcription

Only a few of an organism’s genes are active or “expressed” at any particular time. Which genes are expressed in a particular cell depends on such factors as the nutrients available, the cell’s state of differentiation, and the cell’s age. There are intricate mechanisms that let the cell regulate the expression of many of its genes. Transcription, the first step in the expres-

introns untranslated portions of genes that interrupt coding regions

Archaea one of three domains of life, a type of cell without a nucleus

sion of the genetic information, is an important point at which gene expression can be regulated.

There are two types of regulation: positive control, in which transcription is enhanced in response to a certain set of conditions; and negative control, in which transcription is repressed. Usually, positive control is used at promoters that are otherwise engaged in the initiation of few RNA molecules. Negative control is used at promoters where many molecules of RNA are initiated.

Activator proteins enable positive control by binding to the promoter to recruit RNA polymerase or other required initiation proteins. Such activator proteins usually bind upstream of the promoter (Figure 1). Increased recruitment then leads to an increased rate of synthesis of RNA for a particular gene. The more regulatory sites that are bound, the greater the increase in the rate of RNA synthesis. Repressor proteins can inhibit initiation of transcription by binding to the promoter and preventing RNA polymerase or a required initiation protein from binding.

In eukaryotes, DNA is “packaged” into **nucleosomes** by being wrapped around **histone** proteins. This can dramatically reduce the ability of genes to be transcribed, because the packaging may hide promoter sequences that are recognized by initiation factors.

Two mechanisms are used to alter the DNA packaging, to regulate transcription. First, enzymes called chromatin remodeling factors can move histone proteins around on the DNA, so that promoter sequences are more accessible or less accessible to the transcription initiation machinery. Second, enzymes can attach small chemical groups, including acetyl, phosphate, methyl or other groups, to the histone proteins. This modification of histone proteins may alter the interaction between the DNA and the histones, or between histones and other proteins, either facilitating or blocking the ability of initiation factors to bind DNA.

Transcription also is regulated by proteins that influence how quickly RNA polymerase moves along the DNA. These proteins, called regulatory elongation factors, may help the polymerase traverse pause sites, and they may facilitate elongation through packaged DNA. On the other hand, they may also facilitate the termination of transcription at specific sites. **SEE ALSO** ARCHAEA; GENE EXPRESSION: OVERVIEW OF CONTROL; NUCLEOTIDE; OPERON; RNA POLYMERASES; RNA PROCESSING; TRANSCRIPTION FACTORS; TRANSLATION.

David T. Auble and Pieter L. de Haseth

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nucleosomes chromosome structural units, consisting of DNA wrapped around histone proteins

histone protein around which DNA winds in the chromosome



enzyme a protein that controls a reaction in a cell

promoter DNA sequence to which RNA polymerase binds to begin transcription

archaeans members of one of three domains of life, have types of cells without a nucleus

eukaryotes organisms with cells possessing a nucleus

eubacteria one of three domains of life, comprising most groups previously

Transcription Factors

Transcription factors are protein complexes that help RNA polymerase bind to DNA. RNA polymerase is the **enzyme** that transcribes genes to make messenger RNA, which is then used to make protein. By controlling RNA polymerase's access to the gene, transcription factors control the rate at which a gene is transcribed. Without transcription factors, cells would not be able to effectively regulate the rate at which genes are expressed.

Basal Transcription Factors Bind to the Gene Promoter Region

Every gene has a region known as the **promoter**. This is a DNA sequence “upstream” from the coding region, to which RNA polymerase must bind before it begins transcribing the coding region of the gene. In eukaryotes, the promoters of many (but not all) genes contain the sequence TATAA twenty-five to thirty nucleotides upstream from the transcription start site (T is the nucleotide adenine; A is thymine). Called the “TATA box,” this sequence binds the TATA-binding protein (TBP), one of the most ancient and most important transcription factors.

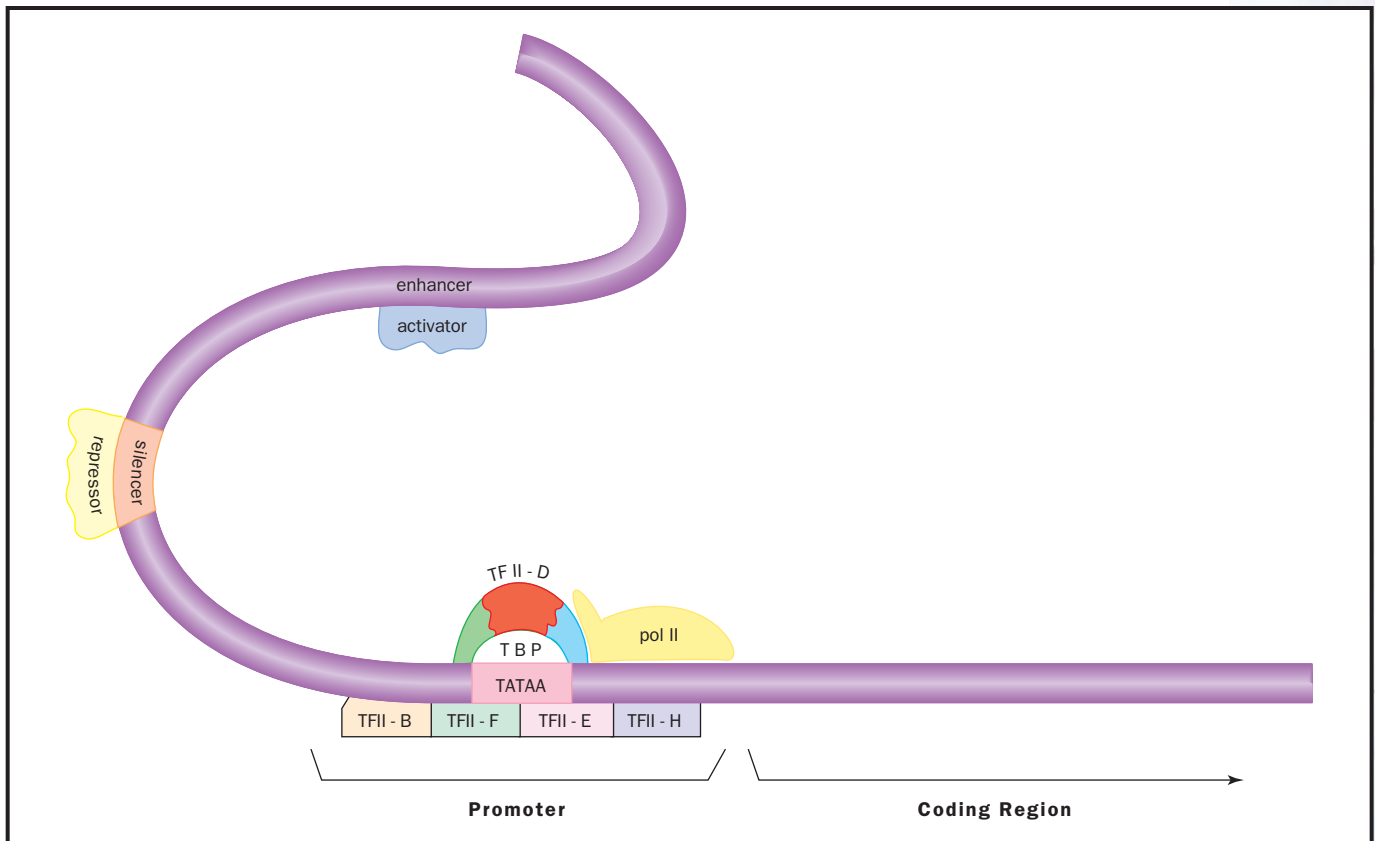
The DNA-binding region of TBP has changed very little in millions of years of evolution, indicating how central this portion of the protein is in gene transcription. Transcription factors in **archaeans** are closely akin to those in **eukaryotes**, though simpler, and they reveal a deep evolutionary relation between the two groups. (Transcription factors are also used by **eubacteria**, but the details differ significantly and will not be discussed here.)

When TBP contacts DNA, the DNA bends. This distortion in shape allows the two sides of the double helix to come apart more easily. TATA-box DNA is especially easy to separate, because successive adenine-thymine pairs are somewhat less stable than series of other nucleotide pairs. The separation of the two strands makes the coding region of the gene more accessible to the RNA polymerase. (There are in fact three eukaryotic RNA polymerases, known as pol I, pol II, and pol III. Each uses a different set of transcription factors; we will discuss those for pol II.)

TBP plays a central role in initiating transcription, but it does not act alone. In archaeans, it works with another protein, transcription factor B. In eukaryotes, TBP is part of a larger complex, TFII-D (this rather colorless name is derived from “transcription factor D for RNA polymerase II”). TFII-D includes several other proteins besides TBP that interact with other factors and help stabilize the assembly on the DNA.

By itself, TFII-D cannot efficiently promote DNA-binding and transcription. Four other factors, TFII-B, -F, -E, and -H (binding in the order listed), allow pol II to bind to the promoter. Together, these are known as the basal transcription factors. Because each of these is composed of numerous individual polypeptides, the entire complex is thought to comprise at least twenty-five interacting polypeptides whose multiple interactions are critical for successful transcription.

The basal transcription factors assembled at the promoter are effective because they bind pol II. To finish their job, though, they must release



it to begin transcription. This occurs when TFII-E and -H cooperate to phosphorylate (add a phosphate group to) RNA polymerase. This changes the polymerase's shape sufficiently to allow it to escape the complex of transcription factors and begin transcription.

Gene-Specific Factors Differentially Enhance Transcription Rates

The basal transcription factors increase the rate of transcription for all genes; indeed, RNA polymerase cannot bind to the promoter without them. However, not all genes should be transcribed at an equal rate all the time. Red blood cells should make lots of hemoglobin but not the digestive enzyme pepsin, while stomach lining cells should do the opposite. Differential control of gene transcription is facilitated by gene-specific transcription factors.

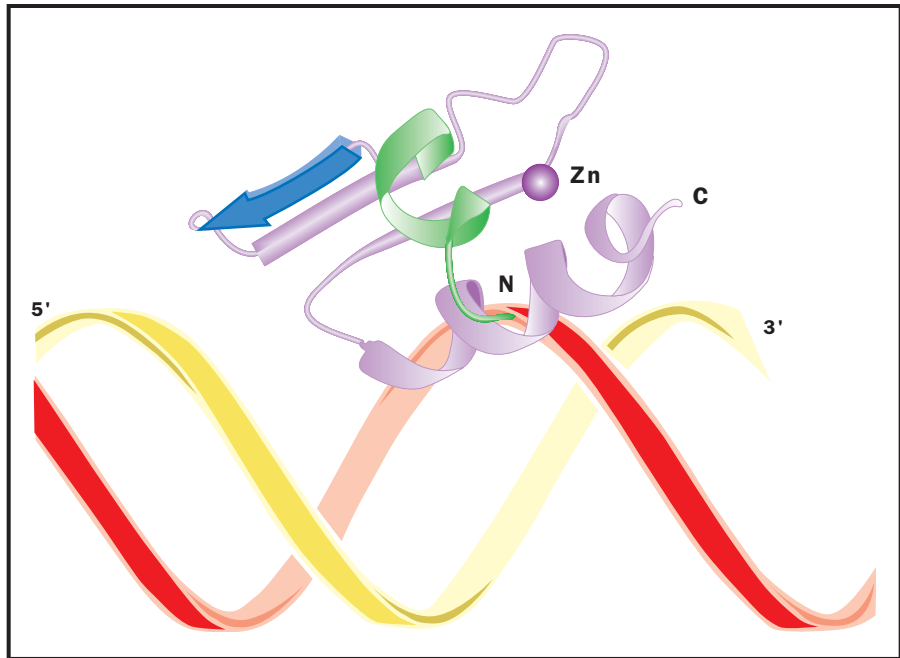
Hormones are an important class of molecules that regulate gene expression. A hormone is not a transcription factor itself but binds to a receptor to form a gene-specific factor. Once bound together, the hormone-receptor complex binds to DNA. Growth factors and homeotic proteins also act as gene-specific factors or form complexes that do.

The number of known gene-specific factors is currently in the low thousands and inevitably will grow as the **genome** becomes better known. An average gene may have several dozen specific factors involved in its regulation, giving the potential for very precise control of its expression.

Schematic diagram of a gene and its regulatory regions. TFII-D, -B, -F, -E, and -H are basal transcription factors, required for binding of RNA polymerase (pol II) to the promoter. Activators and repressors enhance or reduce the rate of gene transcription.

genome the total genetic material in a cell or organism

The zinc finger domain of a transcription factor binds to DNA. The rest of the protein, not shown, is considerably larger. The zinc atom binds to two amino acids in the upper loop, and two others in the lower helix. Adapted from Dutnall, 1996.



Gene-specific factors are known as activators or repressors, depending on whether they increase or decrease the rate of transcription. The DNA sequences that activators bind to are called enhancer sites; repressors bind to silencer sites. Since enhancer and silencer sites are on the same DNA sequence as the gene they control, they are called “cis” regulatory elements (from the Latin word for “side”). The factors that bind to them come from elsewhere in the genome and are called “trans” acting factors.

Many gene-specific factors bind to the promoter outside of the TATA box, especially near the transcription initiation site, the beginning of the DNA sequence that is actually read by RNA polymerase. Others bind to sequences within the coding region of the gene, or downstream from it at the termination region. Some bind to DNA sequences hundreds or thousands of nucleotides away from the promoter. Because of the looped structure of DNA, these sequences are physically close to the promoter, despite being far away along the double helix. The binding sites of transcription factors can be determined by “DNA footprinting.”

Gene-specific factors work in a variety of ways. Some interact with the basal factors, altering the rate at which they bind to the promoter. Some influence RNA polymerase’s rate of escape from the promoter, or its return to it for another round of transcription.

Some factors physically alter the local structure of the DNA, making it more or less accessible. In **eukaryotic** organisms, DNA is wound around protein complexes called **histones** and is further looped, coiled, and condensed to allow efficient packing in the cell nucleus. This arrangement keeps the DNA well ordered but also decreases its accessibility for transcription. By interacting directly with DNA, transcription factors can open up otherwise inaccessible regions.

eukaryotic describing an organism that has cells containing nuclei

histones proteins around which DNA winds in the chromosome

Gene-Specific Factors Share Several Common DNA-Binding Motifs

Gene-specific factors must position themselves on specific DNA sequences to exert their effects, and the relatively simple structure of DNA offers only a few ways for proteins to grab on. Therefore, despite the wealth of different individual transcription factors, each employs one of only a handful of structural “motifs” to bind to the DNA double helix. Each motif is a small portion of a much larger protein, whose other portions confer DNA-sequence specificity and control its interaction with the basal factors or other proteins.

The helix-turn-helix motif is composed of a short section of alpha-helix, linked to a loop of amino acids that changes the direction of the chain, followed by another alpha-helix. The first helix fits into the so-called major groove of the DNA double helix. The side chains of the protein’s amino acids make contact with the exposed portions of the nucleotides. The shape and charges of the one complement those of the other, allowing them to bind; this provides the sequence-specificity needed for effective gene regulation. The homeotic proteins are a special class of proteins employing a modified helix-turn-helix motif. These proteins play critical roles in regulating development in organisms as diverse as fruit flies and humans.

The zinc-finger motif is constructed around an atom of zinc, which binds four amino acids to hold the amino acid chain in proper orientation. While many of the other amino acids vary among different types of zinc-finger proteins, the four key amino acids—either four cysteines or two cysteines and two histidines—are invariant in this class of transcription factors. This group of factors includes the steroid receptors. Steroids are a class of **hormones**, including testosterone and the estrogen, that exert profound effects on development. Steroids must bind to a receptor to form the transcription factor complex. **Mutations** in steroid receptors are responsible for a large variety of inherited disorders, including androgen insensitivity syndrome, thyroid hormone resistance syndrome, and some forms of prostate cancer, breast cancer, and osteoporosis.

Regulation of Transcription Factors

All cells need to be responsive to their environments, whether that environment is the pond-water habitat of a *Paramecium* or the thousands of other cells that a single neuron communicates with every second. Transcription factors are a central feature of this responsiveness. The hormone-receptor complex mentioned above provides a model for understanding how a cell can coordinate its gene expression with external events. The hormone acts as a signal that a change has occurred in the outside world that requires action by the cell, whether it be to grow or divide, or to release its own hormone.

A key feature governing a cell’s repertoire of responses is the set of receptors it makes. Cells that should not respond to testosterone need only ensure that they do not make the testosterone receptor—a decision itself governed by the presence or absence of other transcription factors. Hormones are not the only type of signal possible. Cells have complex networks of signaling pathways that help to regulate their actions.

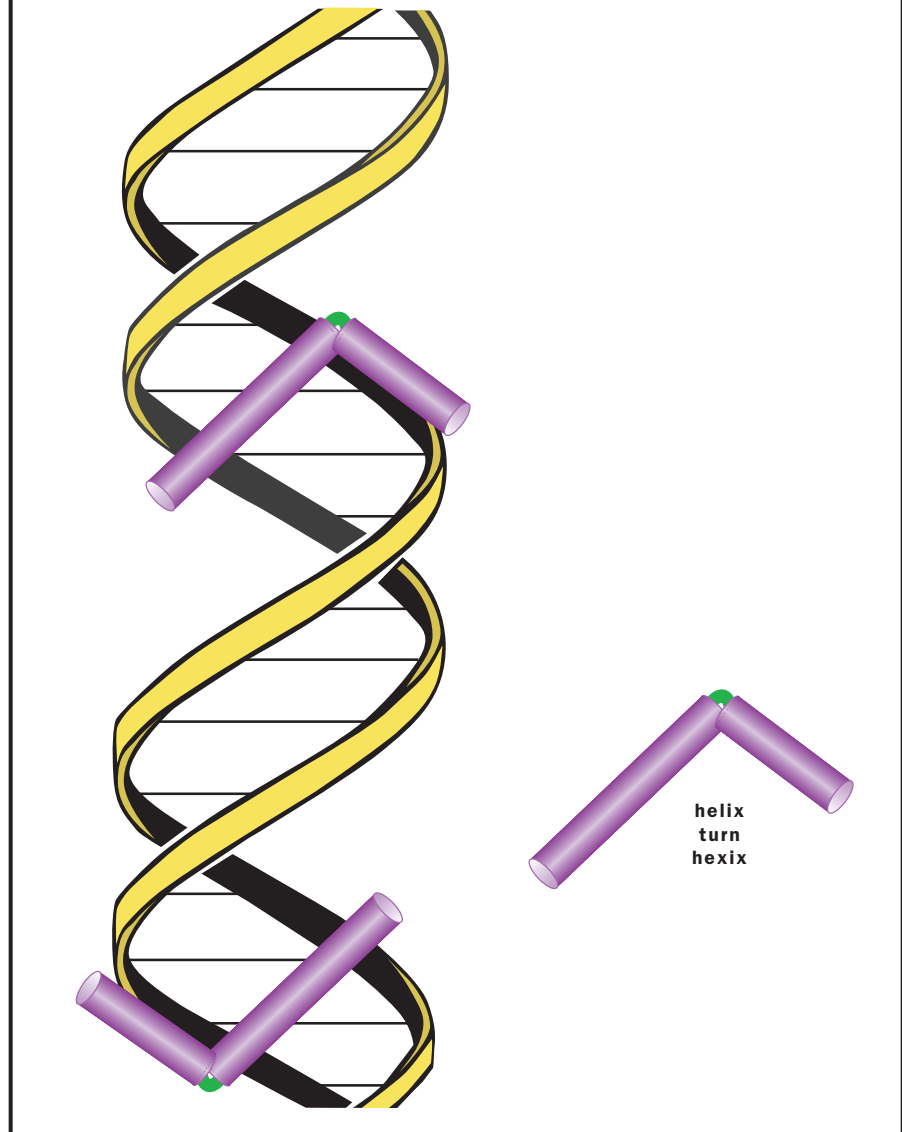
hormones molecules released by one cell to influence another

mutations changes in DNA sequences



A helix-turn-helix motif. Note how one of the two helices (barrels) fits into the major groove of the DNA double helix. The rest of the protein (not shown) aids in binding to the DNA, and interacting with other factors. Adapted from <http://www.micro.iastate.edu/~micro320/Lecture23/hth.html>.

BINDING MOTIF FOR DNA



The exquisite coordination of cellular processes needed to maintain life might be likened to a symphony, in which the many different instruments must play their parts in time with all the others. The timing of gene expression is one of the great puzzles of understanding life: How does each gene get turned on and off at the right time? Although transcription factors are clearly an important part of the answer, they themselves are proteins—the products of genes that must be regulated by yet other transcription factors.

The way out of this paradox is to remember that each organism does not arise from nothing, nor does it spring forth fully formed, with all of its parts fully functioning. Rather, it develops from a preexisting cell, with a specific set of transcription factors in place to turn on a specific set of developmental genes, many of which are themselves transcription factors that

turn on other genes. Although this is only the barest outline of an explanation that is still being worked out, it is clear that the pulsing interplay of transcription factors is a central feature of life's coordinated complexity. SEE ALSO DEVELOPMENT, GENETIC CONTROL OF; DNA FOOTPRINTING; GENE EXPRESSION: OVERVIEW OF CONTROL; HORMONAL REGULATION; PROTEINS; RNA POLYMERASES; SIGNAL TRANSDUCTION; TRANSCRIPTION.

Richard Robinson

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Transduction

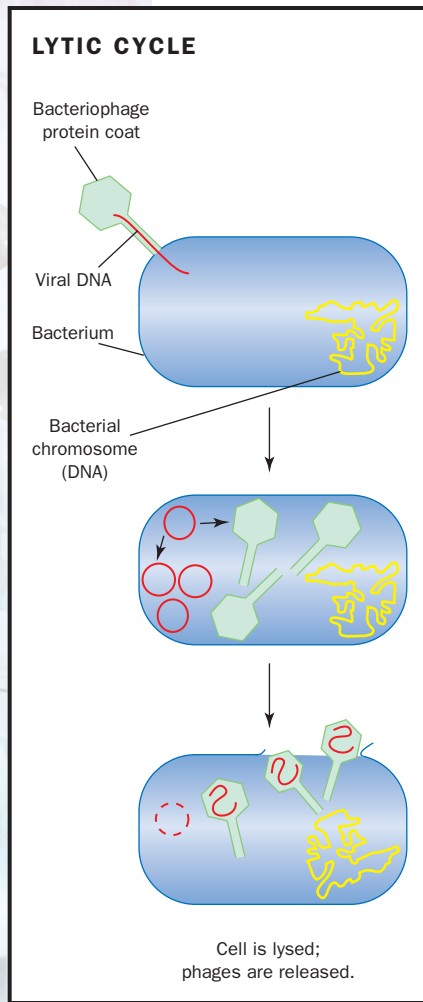
Transduction is one of three basic mechanisms for genetic exchange in bacteria. Like transformation and conjugation, transduction allows the movement of genetic information from a donor cell to a recipient. Unlike the other mechanisms, however, transduction requires the participation of a type of virus called a bacteriophage in order to accomplish this movement. While transduction has been studied in the laboratory since the 1950s, more recently scientists have shown that the process also occurs in nature and probably plays an important role in the evolution of bacteria.

While transduction is common to many bacteria (but not all), the processes can be divided into two basic mechanisms. Generalized transduction tends to transfer all bacterial genes with similar frequencies, or number of cells genetically altered as a function of the total number of potential recipient cells. Specialized transduction tends to transfer only specific genes. It is the life cycle of the particular virus involved in transduction that determines which mechanism will occur, because the transduction of bacterial chromosomal genes is, in fact, the result of an error in the mechanism for viral replication. Thus, to understand the processes of transduction, one must understand the basic mechanisms of viral replication.

Generalized Transduction

Generalized transduction is usually mediated by certain lytic viruses. A lytic virus is one that normally infects a host cell and redirects the resources of the host away from its own cellular replication toward viral replication and the eventual lysis, or breaking open, of the bacterial cell. Typical lytic viruses





In transduction, a virus inserts its genetic material into a bacterial cell, as shown, but then integrates it into the bacterial chromosome. When the viral DNA leaves the bacterial chromosome, it may take one or more bacterial genes with it, packaging them into viral protein coats in which they are released from the cell.

transcription messenger RNA formation from a DNA sequence

translation synthesis of protein using mRNA code

nuclease enzyme that cuts DNA or RNA

use the **transcription** and **translation** mechanisms of the host cell to express viral genes.

Viral genes are expressed in a specific order. First, the genes needed to hijack host cell metabolism are expressed. Next, the viral genetic information is copied many times. The third step involves transcription and translation of viral genes for protein components. The fourth step requires the insertion of viral DNA into the capsid, the protein shell of the virus. Finally, the virus expresses genes needed to break open the host cell, releasing hundreds of new viral particles that move on to infect new host cells.

Among the best studied of the generalized transducing viruses is P22, a virus of the bacterium *Salmonella choleraesuis* (subsp. *typhimurium*). In the course of its normal life cycle, the virus binds to a specific receptor on the surface of the bacterial cell, injects the viral DNA into the host cell, and begins to express viral genes. The ends of the viral DNA are redundant; that is, they have small sections in which the genetic information is duplicated. The DNA repair enzymes of the host cell joins these ends so that the viral DNA forms a circle of double-stranded DNA. A DNA **nuclease** enzyme then makes a nick in one of the strands of DNA. The unbroken strand then serves as a template, allowing the broken strand to extend itself. This forms a long stretch of DNA that repeats the genome of the virus many times.

The long repeating sequence of DNA is known as a concatamer. The genes for viral capsid proteins (the proteins that make up the coat of the virus) are then expressed, and empty viral capsids are produced. Next, a viral DNA-cutting enzyme cuts the concatamer of DNA at a specific sequence. Once the concatamer is cut, the end of the DNA is pushed into an empty viral capsid until the viral head is full. A new empty capsid then positions itself on the end of the concatamer and the process is repeated until all the viral DNA is packaged.

Generalized transduction occurs when the enzyme that normally cuts the concatamer cuts the host chromosome instead. The viral capsids cannot distinguish viral DNA from host DNA; thus, once the initial cut is made, the empty capsids are filled with host chromosomal DNA instead of viral DNA. Since each empty capsid picks up where the last ended, it is possible for all the genes on the host chromosome to be packaged at similar rates: thus, this process is generalized. When the host cell breaks open, the viral capsids containing host chromosomal DNA, now called transducing particles, are able to bind to a new host cell. The DNA they carry is injected into the new host, but since there are no viral genes included on the injected DNA, the newly infected host cell is not killed. If the original host cell (the donor) was genetically different from the new host cell (the recipient), the DNA recombination enzymes of the recipient cell will insert the new genes in place of the old, thus altering its own genetic makeup. This genetic change is integrated into the recipient genome and is passed on to future progeny cells.

Specialized Transduction

Specialized transduction results in the movement of only specific genes. The viruses that carry out specialized transduction are called lysogenic viruses. They have a mechanism for replication that is different from that of generalized transduction, for they integrate their DNA directly into the

chromosome of its host's genome. Each time the host chromosome is duplicated, so is the integrated viral DNA. In many cases these viruses express genes that keep the viral DNA dormant; that is, the virus does not immediately replicate.

The best characterized model for a specialized transducing virus is phage Lambda of *Escherichia coli*. The Lambda virus begins its life cycle in much the same way as P22. A tail fiber on the end of the virus specifically binds to a receptor, called the maltose binding protein, on the surface of the *E. coli* cell. The viral DNA is then injected into the host cell. On the chromosome of the virus is a section of DNA that is almost identical to a DNA sequence found on the bacterial chromosome. The recombination enzymes of the *E. coli* host break and rejoin the viral DNA and host DNA together at this site, thus integrating the virus genome into the chromosome of the host bacterium.

A viral gene, the repressor gene, is then expressed and keeps the virus from activating its own replication. This integrated virus, called a prophage, can be maintained stably as a part of the host chromosome as long as the host cell remains healthy. If the host cell becomes damaged, enzymes are activated that destroy the Lambda repressor protein. Without this protein, the viral DNA will break out of the host chromosome and begin to replicate itself, much as a lytic virus does. Ultimately, the virus particles are packaged, released, and move on to infect a new host cell.

Specialized transduction occurs when the enzyme that cuts the viral DNA out of the host chromosome makes a mistake and cuts in the wrong place, removing some, but not all, of the viral genes. Since Lambda capsids fill by a “headful” mechanism, small bits of the host chromosome are packaged along with part of the viral genes. These viral particles are called defective particles because some of the viral genes are missing in the package and thus, when the virus infects a new host cell, not all the genes needed for viral replication are present. Lacking the ability to replicate, the virus cannot kill its new host. Because of this mechanism of viral packaging, specialized transducing viruses can pick up genes only on either side of the site where the virus integrates into the bacterial chromosome. Thus, while generalized transducing viruses can move any genes, specialized transducing viruses move only specific genes.

Uses in Research

Generalized transducing viruses are the most useful in mapping bacterial chromosomal genes. Since the amount of DNA that is packaged by the virus is determined by the size of the head of the virus, each viral particle holds the same amount of DNA. The initial cutting of the host chromosome is a random event, giving all genes approximately the same probability of being packaged and transferred. Each piece of DNA that is packaged will be the same length, meaning that the closer together two genes are, the higher the probability that the two genes will be present on the same fragment of packaged DNA. In other words, the closer together the genetic markers are, the higher the frequency of cotransduction. Therefore the distance between closely linked chromosomal genes can be calculated by measuring the frequency that two genes or genetic markers are cotransduced.



When the distance between two genes is greater than the size of the viral genome, it is physically impossible for the two genes to be packaged in the same viral capsid. Thus, these genes are said to be “unlinked” with regard to viral mapping. Since most transducing viral capsids can hold only from about fifteen to fifty genes, transductional mapping of bacterial chromosomal genes is most effective for genes that are relatively close to one another.

Historically, viruses, including transducing viruses, have played an important role in defining the basic principles of molecular biology. Perhaps the most important contribution to the study of transduction was that made by Alfred Hershey and Martha Chase. During the 1940s and 1950s there was still a great deal of controversy over whether DNA or protein was the genetically inheritable material. Hershey and Chase recognized that the simplicity of the virus, consisting of DNA wrapped in a protein coat, was the ideal model to directly address the question of the basis for inheritance.

They began their experiments by growing viruses on host bacteria in **media** containing radioactive forms of sulfur and phosphorus. The radioactive sulfur labeled the protein components of the virus, while the radioactive phosphorus labeled the DNA portions. This allowed them to independently track the protein and DNA. After separating the radioactively labeled viruses from their host cells, they used the viruses to infect host bacteria that were not radioactively labeled. After infection, they separated the bacterial cells from the growth media. Radioactive phosphorus (viral DNA) was found inside the host cells, while the radioactive sulfur (viral proteins) was found outside the cell. This indicated that only the DNA of the virus enters the host cell, while the protein was left on the outside.

Further details of their experiment make the case for DNA even more strongly. When viruses are grown on bacteria in a thin layer on the surface of agar plates, each virus will create a clear spot called a plaque, indicating infection. Hershey and Chase had two different mutants of their virus that resulted in plaques that looked different from the normal viral infection.

When the infected bacteria were replated, normal plaques were seen, indicating that the two different new mutants had both infected the same host cell and that recombination between the virus DNA occurred within, making a virus that had repaired both mutations. Consequently, since only DNA had entered the host cells and genetic change had occurred in the viruses, DNA had to be the inheritable material. Proteins could not be the source of inheritance because the viral proteins never entered the host cells. **SEE ALSO** CONJUGATION; DNA REPAIR; *ESCHERICHIA COLI* (*E. COLI* BACTERIUM); EUBACTERIA; MAPPING; NATURE OF THE GENE, HISTORY; RECOMBINANT DNA; TRANSFORMATION; VIRUS.

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media (bacteria) nutrient source

Transformation

Transformation is one of three basic mechanisms for genetic exchange in bacteria. Transformation may be either a natural process—that is, one that has evolved in certain bacteria—or it may be an artificial process whereby the recipient cells are forced to take up DNA by a physical, chemical, or enzymatic treatment. In both cases, **exogenous** DNA (DNA that is outside the host cell), is taken into a recipient cell where it is incorporated into the recipient **genome**, changing the genetic makeup of the bacterium.

Natural Transformation

Natural transformation is a physiological process that is genetically encoded in a wide range of bacteria. Most bacteria must shift their physiology in order to transform DNA; that is, they must become “competent” for taking up exogenous DNA. There appear to be two basic mechanisms by which bacteria can become competent for transformation. In some bacteria, including *Streptococcus pneumoniae* and *Bacillus subtilis*, competence is externally regulated. These bacteria produce and secrete a small protein called competence factor that accumulates in the growth medium.

When the bacterial culture reaches a sufficient density, the concentration of competence factor reaches a level high enough to bind receptors on the outside of the cell. This event causes an internal signal to turn on the expression of the genes needed for transformation. Thus, competence development is controlled by cell density. There are a number of other bacterial functions that are similarly regulated, and these processes are collectively called quorum sensing mechanisms. In other bacteria, including *Haemophilus influenzae* and *Pseudomonas stutzeri*, competence development is internally regulated. When there is a shift in the growth dynamics of the bacterium, an internal signal triggers competence development.

Once competence is induced, three additional steps are required for natural transformation. After induction of competence, double-stranded DNA is bound to specific receptors on the surface of the competent cells. These receptors are lacking in noncompetent cells. The double-stranded DNA is nicked and one strand is degraded while the other strand enters the cell. This process is called DNA uptake. Finally, the recombination enzymes of the recipient cell will bind the single-strand DNA that has entered it, align it with its **homologous** DNA on the recipient chromosome, and recombine the new DNA into the chromosome, incorporating any genetic differences that exist on the entering DNA.

Artificial Transformation

While a wide variety of bacteria can transform naturally, many species cannot take up DNA from an outside source. In some cases DNA can be forced into these cells by chemical, physical, or enzymatic treatment. This is especially important in genetic engineering, as artificial transformation is essential for the introduction of genetically altered sequences into recipient cells. One of the two most common methods is a chemical process where cells are heat-shocked, then treated with the DNA and a high concentration of calcium ions. The calcium ions precipitate the DNA on the surface of the cell, where the DNA is forced into the recipient.

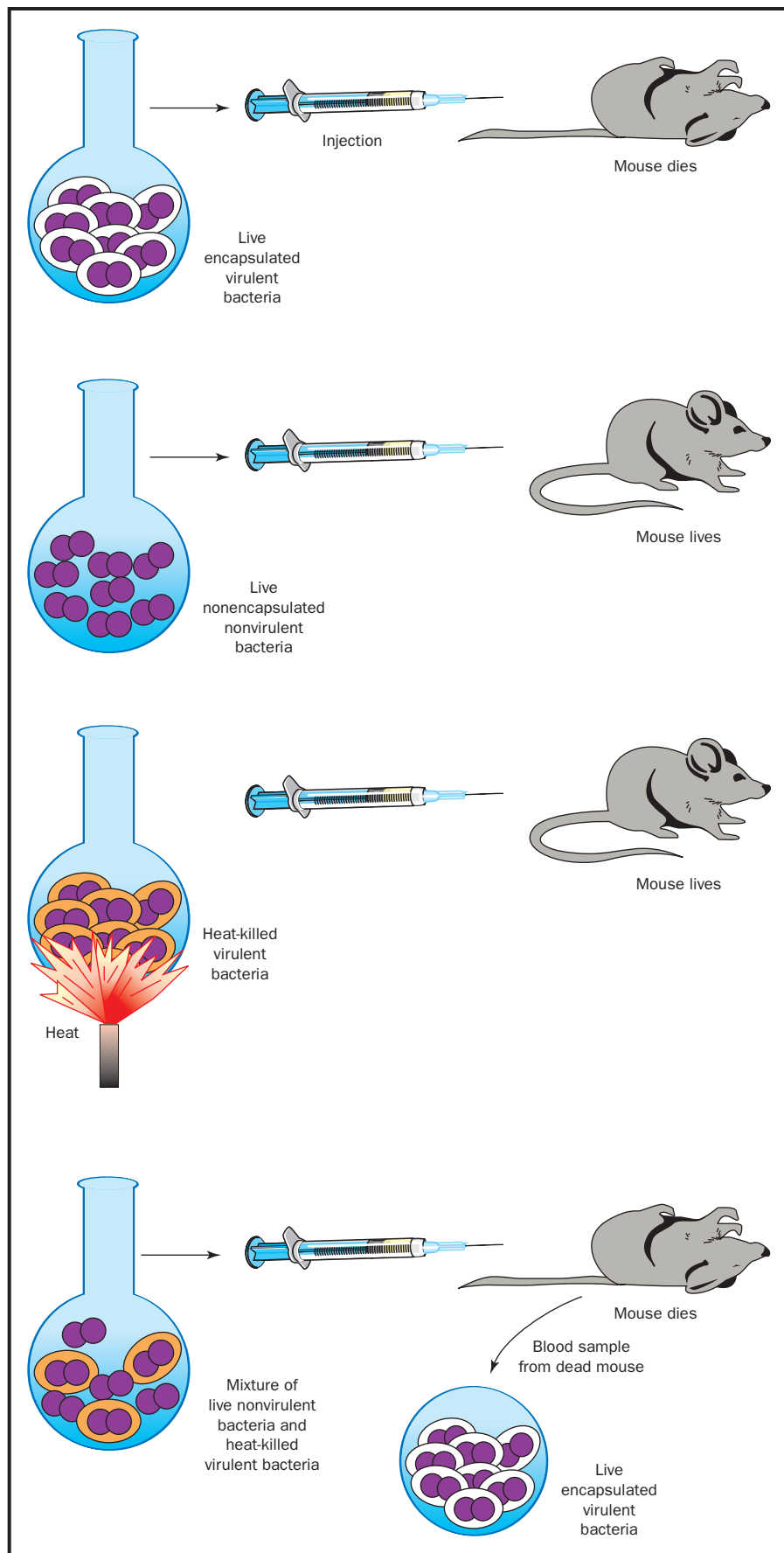
exogenous from outside

genome the total genetic material in a cell or organism

homologous containing the same set of genes



In transformation, bacteria pick up DNA from their environment. This illustration shows Frederick Griffith's classic experiment which first demonstrated transformation. The live nonvirulent bacteria absorb DNA from the dead nonvirulent bacteria, and become virulent themselves. Adapted from Curtis and Barnes, 1994.



More recently a new method, called electroporation, has been used to introduce DNA by artificial transformation. In this process a suspension of recipient bacteria and transforming DNA is placed in a container with metal sides. A high-voltage electrical current is passed through the sample, temporarily creating small pores, or channels, in the membranes of the bacteria. The DNA enters the cells and the pores close. Thus, exogenous (outside) DNA is introduced into the recipient.

Because exogenous DNA is not enclosed within cell walls, it is susceptible to enzymes that degrade DNA, called DNases. A hallmark of transformation is that it is sensitive to DNase, while the other two processes of genetic exchange, **transduction** and **conjugation**, are DNase resistant. Transduction is DNase resistant because the DNA is protected inside a viral protein coat. Conjugation is DNase resistant because fusion occurs between donor and recipient cells, meaning the DNA is never exposed to the outside environment or to enzymes.

transduction conversion of a signal of one type into another type

conjugation a type of DNA exchange between bacteria

Discovery of Transformation

The first report of transformation was an example of natural transformation. Dr. Frederick Griffith was a public health microbiologist studying bacterial pneumonia during the 1920s. He discovered that when he first isolated bacteria from the lungs of animals with pneumonia, the bacterial colonies that grew on the agar plates were of reasonable size and had a glistening, **mucoïd** appearance. When he transferred these colonies repeatedly from one agar plate to another, however, mutant colonies would appear that were much smaller and were chalky in appearance. He designated the original strains as “smooth” strains, and the mutants as “rough” strains. When Griffith injected mice with smooth strains they contracted pneumonia, and smooth strains of the bacterium could be reisolated from the infected mice. However, when he infected the mice with rough strains they did not develop the disease. The smooth strains were capable of causing disease, or were “virulent,” while the rough strains did not cause disease, or were “avirulent.”

mucoïd having the properties of mucous

Griffith questioned whether the ability to cause disease was a direct result of whatever product was making the bacterial colonies smooth, or whether rough strains of the bacterium were less capable of establishing disease for some other reason. To investigate this idea, he prepared cultures of both bacterial types. He pasteurized (killed) each of these cultures by heating them for an hour and then injected the heat-treated extracts into mice. His hypothesis was that if the bacteria had to be living to cause disease, heat-treating that killed the bacteria would prevent disease. If, on the other hand, the smooth material was itself a toxin, heating would not destroy it, meaning heated extracts of smooth strains would continue to cause disease. When Griffith injected heated extracts of both smooth and rough strains into mice, neither caused disease. This suggested to him that only living smooth cells could cause disease.

In his next experiment he coinjected unheated, live rough bacteria with heat-treated, dead smooth bacteria into mice. All of the mice developed disease, and when bacteria were isolated from the lungs of the diseased mice, all the isolates were smooth. This led Griffith to propose that there was some “transforming principle” in the heated smooth extract that con-



verted the rough strains back to smooth ones capable of causing diseases. Griffith was not able to determine the nature of this transforming principle, but his experiments suggested that some “inheritable” material present in the heated extract could genetically convert strains from one colony type to another.

Approximately ten years later, another research team, that of Oswald Avery, Colin Munro MacLeod, and Maclyn McCarty, followed up on Griffith's experiments by enzymatically and biochemically characterizing the heated transforming extracts that Griffith had produced. Their studies indicated that the transforming principle was deoxyribonucleic acid (DNA), providing the first definitive evidence that DNA was the inheritable material. SEE ALSO CONJUGATION; NATURE OF THE GENE, HISTORY; RECOMBINANT DNA; TRANSDUCTION.

Gregory Stewart

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Transgenic Animals

The term “transgenics” refers to the science of inserting a foreign gene into an organism's **genome**. Scientists do this, creating a “transgenic” organism, to study the function of the introduced gene and to identify genetic elements that determine which tissue and at what stage of an organism's development a gene is normally turned on. Transgenic animals have also been created to produce large quantities of useful proteins and to model human disease.

In the early 1980s Frank Ruddle and his colleagues created the first transgenic animal, a transgenic mouse. Researchers making transgenic mice use a very fine glass needle to inject pieces of DNA into a fertilized mouse egg. They inject the DNA into one of the egg's two **pronuclei**, before the pronuclei fuse to become the nucleus of the developing embryo's first cell. After the DNA is injected, multiple copies, usually joined end-to-end, insert randomly into the host organism's nuclear DNA.

Multiple injected embryos are then transferred to a surrogate mother mouse to develop to term. Only a small percentage of the embryos survive the injection, and even of those that survive, not all have successfully incorporated the foreign DNA into their genome. Once the mice are born, researchers must identify which mice have the foreign gene in their genome. The animals that contain the added foreign DNA, or transgene are referred to as transgenics.

genome the total genetic material in a cell or organism

pronuclei egg and sperm nuclei before they fuse during fertilization

Targeted Gene Replacement and “Knockouts”

A gene that is injected into a fertilized egg is generally integrated randomly into the host genome. This means that scientists originally had no control over where in the host genome the foreign gene would land, nor could they control the number of copies of the gene that would be integrated. Where and how many copies of a gene are inserted can profoundly affect its function, so scientists looked for ways to make more precise insertions.

In the late 1980s Mario Capecchi and colleagues pioneered a method to target the inserted gene to a desired position in the genome. These researchers took advantage of an observation that, on rare occasions, an injected, mutated copy of a gene lines up precisely with the original form of the gene in the mouse genome. By a process called homologous recombination, the aligned DNA segments are cut and rejoined to each other. The result is a precise stitching of the introduced DNA into the targeted gene in the mouse genome. This means that scientists found they could make minor modifications to a gene before injecting it and, by homologous recombination, or “gene targeting,” replace the natural gene with this transgenic version. For commercial applications, the transgene is often a gene coding for a functional human protein, which is then mass-produced in the host organism and isolated. For research purposes, it is often more useful to insert a mutated, nonfunctional version of the gene, to see what happens when the normal, functional version is missing. Creating such “knockout” organisms is a key tool used for studying genes that control development.

Selection of Gene Targeted Cells

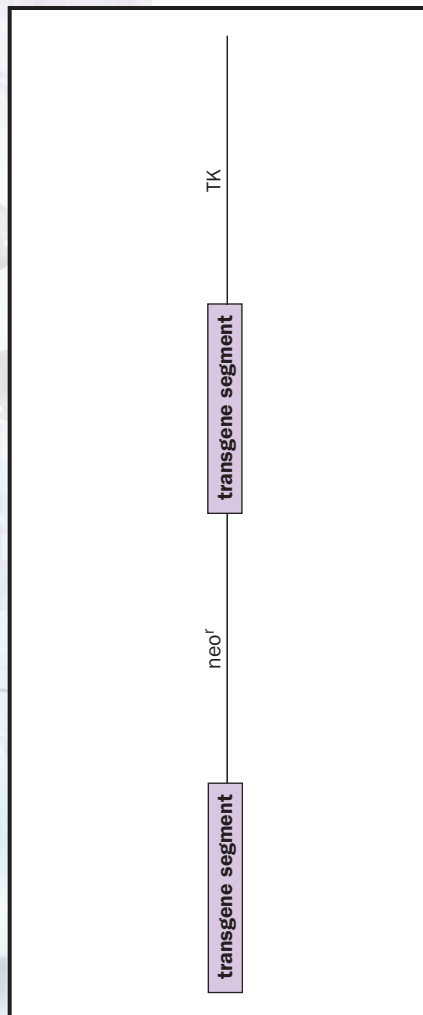
Homologous recombination is a very rare event, and scientists using it to modify or “knock out” mouse genes must identify the cells in which it has occurred. In addition to injecting the gene they are trying to incorporate, scientists also inject “selectable” genes whose products permit cells to live or cause them to die in the presence of a particular drug. The two most common selectable genes used in gene targeting are the neomycin resistance (*neo^r*) gene, which allows cells to survive in the presence of the antibiotic neomycin, G418, and the thymidine kinase (*TK*) gene from the herpes virus. Cells with this gene die in the presence of the antiviral agent gancyclovir. The *neo^r* and *TK* genes are generally used together for maximum selection.

The first step in gene targeting is to clone the gene that is to be replaced from the mouse genome. The cloned gene is placed into a targeting vector along with a selectable gene such as the *neo^r* gene. (The targeting vector is a larger piece of carrier DNA.) When the targeting vector lines up with the native mouse gene and homologous recombination stitches the genes in the targeting vector into the genome, the *neo^r* gene will be included. By adding G418 to the cell growth media, only those cells that have incorporated into their genome the transgenes, including the *neo^r* gene, will survive. This is referred to as positive selection, selecting for cells that contain the desired integration product.

The *TK* gene is placed in the same targeting vector, but it is placed outside of the cloned mouse gene pieces. If homologous recombination occurs such that the added DNA lines up precisely with the native gene, the *TK* gene will be excluded. However, if the targeting vector integrates randomly



Chimera generated by injecting RW4 ES cells. Black mouse is a littermate control.



Targeting vector.

blastocyst early stage of embryonic development

into the genome the entire vector is inserted and the *TK* gene will be included. The addition of gancyclovir kills all cells that have the *TK* gene, thereby killing those where the insertion was random. The combination of positive *neo^r* selection and negative *TK* selection results in the survival of only those cells containing targeted gene replacements.

From Transgene to Transgenic Organism

To get the targeted gene into the mouse genome, Capecchi used a very specialized embryonic cell previously isolated by Matthew Kaufman and Martin Evans. These embryonic stem (ES) cells were isolated from an early stage of embryonic development (the **blastocyst**). When grown under controlled conditions in culture dishes, the ES cells have the remarkable capacity to become cells belonging to any tissue type. They can become muscle, cartilage, blood vessel or nerve cells, for example. Even more astonishing, when ES cells are injected back into a blastocyst, they mix with the cells of the recipient embryo and contribute some cells to every tissue in the body. Thus, if researchers place a transgene into the ES cell's genome and inject those ES cells into a blastocyst, the transgene could end up in every tissue type of the mouse.

Typically, the targeting vector is placed into ES cells derived from a mouse that has a brown coat. ES cells with the targeted gene replacement are identified through positive and negative selection. They are then injected into blastocysts from mating mice with black coats. These blastocysts are then placed into a surrogate mother and allowed to develop to term. The offspring that have incorporated transgene-containing ES cells into some of their tissues are identified by having patches of brown coat color.

To allow the targeted gene replacement to be passed to subsequent generations, the ES cells must also contribute to the developing embryo's eggs or sperm. To determine if the targeted gene has been incorporated into a mouse's eggs or sperm, the mice with brown patches are mated to black-coated individuals. The brown coat color is a dominant trait, so any offspring with brown coats can be assumed to have arisen from germ cells that derived from the manipulated ES cells and thus contain the targeted gene replacement. These mice are mated to brown-coated siblings to produce homozygous transgenics, which are identified by determining if the offspring contain two copies of the transgene replacing both normal copies of the gene in the genome. These homozygous mutants are studied to look for phenotypic changes due to the transgene.

Applications

Gene targeting has been used to identify the function of hundreds of mouse genes. One dramatic example was the deletion of the *Lim-1* gene by Richard Behringer and colleagues. The mice carrying this deletion died during embryonic development because of a complete lack of brain and head structure development. This demonstrated that the *Lim-1* gene was critical for head development.

The gene responsible for sex determination in mice was also identified thanks to the use of transgenic animals. When a gene called *Sry* (sex-determining region of the Y chromosome) was microinjected into mouse

embryos, the resulting transgenic mice were all male. Indeed, even in the mice that had two X chromosomes and were thus genetically female, the presence of the *Sry* gene was sufficient to cause them to develop testes and led to complete sex reversal. This clearly demonstrated that the *Sry* gene alone was responsible for sex determination.

Gene targeting is also being exploited by scientists to create models of human disease. For instance, **mutations** have been made in the mouse version of the cystic fibrosis transmembrane conductance regulator gene. Although mice with the mutated gene do not develop the devastating symptoms of cystic fibrosis in their lungs, they do develop the intestinal and pancreatic duct defects associated with the disease and thereby provide a model to study at least part of the disease. Transgenic mice overexpressing the amyloid precursor protein form deposits in the brain that resemble the amyloid plaques found in Alzheimer's patients. Mouse models such as these can potentially be used to test drug therapies and to learn more about the progression of the disease.

One important application of transgenic technology is the generation of transgenic livestock as "bioreactors." Key human genes have been introduced into sheep, cows, goats, and pigs so that the human protein is secreted into the milk of the transgenic animal. In theory, large quantities of the human protein can be produced in the animal's milk and subsequently purified for use in medical therapies. An early example of this technology by John Clark and colleagues was the production of transgenic sheep expressing the human blood-clotting factor IX needed by many patients with hemophilia. These researchers placed the human factor IX gene under the control of a piece of sheep DNA that normally turns on the beta-lactoglobulin gene in the mammary tissue. Though the sheep secreted factor IX into their milk, the levels of the protein were very small. With advances in the efficiency of creating and expressing genes in transgenic farm animals, therapeutic proteins can now be isolated. **SEE ALSO** AGRICULTURAL BIOTECHNOLOGY; ALZHEIMER'S DISEASE; BIOTECHNOLOGY; CLONING GENES; CLONING ORGANISMS; CYSTIC FIBROSIS; EMBRYONIC STEM CELLS; GENE TARGETING; HEMOPHILIA; RECOMBINANT DNA; RODENT MODELS; SEX DETERMINATION; TRANSGENIC MICROORGANISMS; TRANSGENIC ORGANISMS; ETHICAL ISSUES; TRANSGENIC PLANTS; Y CHROMOSOME.

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Transgenic Microorganisms

A transgenic microorganism is a microbe, usually a bacterium, into which genetic information has been introduced from the outside and which possesses the ability to pass that information on to subsequent generations in a

mutations changes in DNA sequences



stable manner. This is not an entirely novel idea in microorganisms, since bacteria have been practicing and perfecting this art over billions of years of evolution. We, on the other hand, have only recently learned to duplicate this phenomenon and turn it to our own purposes. Genetic engineering is the field that has developed as a consequence of research into this process. Its commercial application forms the basis of the biotechnology industry today.

Moving Genes between Species

The process by which scientists introduce new genetic material into a microorganism is called molecular or gene cloning. It involves the isolation of DNA from a source other than the microorganism itself. Source organisms span the world of living things, from microbes to plants to animals, including humans. Scientists obtain source DNA in several different ways: by disrupting cells of the target microbe (or plant or animal) and fragmenting it into small pieces, by synthesizing it from an RNA template using an enzyme called reverse transcriptase, or by knowing the specific gene sequence and synthesizing it directly in the laboratory.

Once obtained, the pieces of DNA are inserted into a small genetic component that has the ability to make copies of itself (replicate) independently from the microbial genome. This self-replicating unit is called a cloning vector. Although these genetic elements exist naturally in the form of **plasmids** and bacterial viruses, many of the ones used today have been altered to improve their properties for transferring genes. **Restriction enzymes**, which nick the donor DNA and the cloning vector at specific sites, and DNA ligase, which attaches the donor DNA to the cloning vector, allow the source genes of interest to be inserted into the cloning vector without disrupting its ability to replicate.

The next step in the process is the introduction of the cloning vector with its segment of new DNA into a living cell. Bacteria have the ability to transport DNA into their cells in a process called transformation, and this ability is commonly exploited to achieve this goal. Getting the DNA into the cell, however, is only the beginning. No transformation is 100 percent efficient, and so the bacteria that receive the gene(s) of interest must be separated from those that did not. One of the best studied and most commonly used cloning vectors, pBR322, is especially useful for this purpose, as it contains several genes for antibiotic resistance. Hence, any cell transformed with DNA containing pBR322 will be antibiotic resistant, and thus can be isolated from similar cells that have not been so transformed by merely growing them in the presence of the appropriate drugs. All that remains is to identify bacteria that are producing the product of the desired gene(s), and cloning is a success.

The introduction of human genes into bacteria has several complicating wrinkles that make cloning them even more challenging. For example, a bacterial gene codes for a protein from start to finish in one long string of **nucleotides**, whereas human cells have stretches of noncoding nucleotides called introns within their genes. Bacteria do not have the same ability as human cells to remove these introns when producing proteins from the gene, and if the introns are not removed, the intended protein cannot be produced. This, along with other complications, has been overcome using many of the tools of genetic engineering.

plasmids small rings of DNA found in many bacteria

restriction enzymes enzymes that cut DNA at a particular sequence

nucleotides the building blocks of RNA or DNA

Commercial Application

Transgenic microbes have many commercial and practical applications, including the production of mammalian products. A company called Genentech was among the earliest and most successful commercial enterprises to use genetically engineered bacteria to produce human proteins. Their first product was human insulin produced by genetically engineered *Escherichia coli*. A variety of other human **hormones**, blood proteins, and immune modulators are now produced in a similar fashion, in addition to vaccines for such infectious agents as hepatitis B virus and measles.

hormones molecules released by one cell to influence another

Another promising application of genetically engineered microbes is in environmental cleanup, or bioremediation. Scientists have discovered many naturally occurring genes that code for enzymes that degrade toxic wastes and wastewater pollutants in bacteria. Examples include genes for degrading chlorinated pesticides, chlorobenzenes, naphthalene, toluene, anilines, and various hydrocarbons. Researchers are using molecular cloning to introduce these genes from several different microbes into a single microbe, creating “super microbes” with the ability to degrade multiple contaminants.

Ananda Chakrabarty created one of the first microbes of this nature in the early 1970s. He introduced genes from several different bacteria into a strain of *Burkholderia cepacia*, giving it the ability to degrade toxic compounds found in petroleum. This microbe offered a potential alternative to skimming and absorbing spilled oil. Chakrabarty’s genetically modified bacterium has never been used, however, due to public concerns about the release of genetically engineered microbes into the environment. The microbe did, on the other hand, play an important role in establishing the biotechnology industry. The U.S. Patent Office granted Chakrabarty the first patent ever for the construction and use of a genetically engineered bacterium. This established a precedent allowing biotechnology companies to protect their “inventions” in the same way chemical and pharmaceutical companies have done in the past. SEE ALSO BIOREMEDIATION; *ESCHERICHIA COLI* (*E. COLI* BACTERIUM); GENE; PLASMID; REVERSE TRANSCRIPTASE; TRANSFORMATION; TRANSGENIC ANIMALS; TRANSGENIC ORGANISMS: ETHICAL ISSUES; TRANSGENIC PLANTS.

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Transgenic Organisms: Ethical Issues

A transgenic organism is a type of genetically modified organism (GMO) that has genetic material from another species that provides a useful trait.



For instance, a plant may be given genetic material that increases its resistance to frost. Another example would be an animal that has been modified with genes that give it the ability to secrete a human protein.

Bioethics addresses the impact of technology on individuals and societies. Bioethical issues include an individual's right to privacy, equality of access to care, and doctor-patient confidentiality. In the case of transgenic organisms, a major bioethical issue is freedom of choice. Yet broader issues also arise, such as the ethics of interfering with nature, and effects of transgenic organisms on the environment.

The changes that are possible with transgenesis transcend what traditional gardening or agriculture can accomplish, although these too interfere with nature. A transgenic tobacco plant emits the glow of a firefly, and a transgenic rabbit given DNA from a human, a sheep, and a salmon secretes a protein hormone that is used to treat bone disorders. If mixing DNA in ways that would not occur in nature is deemed wrong, then transgenesis is unethical. Said a representative of a group opposed to GMOs in New Zealand at a government hearing, "To interfere with another life-form is disrespectful and another form of cultural arrogance."

A more practical objection to transgenic technology is the risk of altering **ecosystems**. Consider genetically modified Atlantic salmon, currently under review at the U.S. Food and Drug Administration (FDA). The fish have a growth hormone gene taken from Chinook salmon and a DNA sequence that controls the gene's expression taken from ocean pout, a fish that produces the hormone year-round. Because Atlantic salmon normally produce growth hormone only during the summer, the transgenic animal grows at more than twice the natural rate. Such genetically modified salmon could escape the farms where they are intended to be raised and invade natural ecosystems, where they may outcompete native fish for space, food, and mates.

Until recently, the fear that a transgenic organism might escape and infiltrate a natural ecosystem was based on theoretical scenarios. For example, a 1999 report of transgenic corn pollen harming Monarch butterfly larvae in a laboratory simulation was not confirmed by larger, more realistic studies. But in 2001 transgenic corn was discovered growing on remote mountaintops in Mexico, ironically in the area where most natural corn variants originated. The corn was not supposed to have been able to spread beyond the fields where it was grown. At about the same time, 10,000 hectares (24,700 acres) of transgenic cotton were found in India. A farmer had crossed transgenic cotton he had obtained from the United States with a local variant and planted crops, not realizing that he had used a genetically modified product.

At the present time, American consumers cannot tell whether a food contains a genetically modified product or not because the two-thirds of processed foods that include GMOs and are sold in the United States have not been labeled. This lack of labeling is consistent with existing regulatory practice. While the FDA tests foods to determine their effect on the human digestive system, their biochemical makeup, and their similarity to existing foods (using a guiding principle called substantial equivalence), foods are not judged solely by their origin. For example, the FDA denied marketing of a potato derived from traditional selective breeding

ecosystem an ecological community and its environment



Another ethical dimension to transgenic organisms is that the methods to create genetically modified seeds, and the seeds themselves, lie in the hands of a few multinational corporations. In the mid-1990s a